General Mathematics — Exercise Sheet 8

Differentiation rules

Variant: Statements

Exercise 1 (A composite with excluded inputs). Find the real domain and derivative of

$$F(x) = \arctan\left(\frac{x}{1-x^2}\right).$$

Simplify the derivative and explain whether that simplified expression assigns a derivative to F at $x = 1$ or $x = -1$. Use Lecture 8's chain, quotient and inverse trigonometric rules.

Exercise 2 (Logarithmic differentiation with two logarithms). For $x > 1$, let $Y(x) = (\ln x)^{\cos x}$. Justify that Y is differentiable and find its derivative by logarithmic differentiation. State why the domain restriction is used.

Exercise 3 (An inverse without an elementary formula). Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = e^x + x$. Prove that f is bijective. For its inverse g, prove differentiability on $\mathbb{R}$, express $g'(y)$ in terms of $g(y)$, and evaluate $g'(2 + \ln 2)$. Use the school order of the exponential, Lecture 6's intermediate value theorem, and Lecture 8's inverse rule.

Exercise 4 (The branch in an inverse hyperbolic composition). Let $H(x) = \operatorname{arcosh}(\cosh x)$ on $\mathbb{R}$. Determine its value and every point where it is differentiable. For nonzero x, compare the chain-rule calculation with the derivative of your simplified function. Use Sheet 2's nonnegative inverse-cosh branch and Lecture 8's hyperbolic formulas.

Exercise 5 (An oscillating function with a continuous derivative). Define $P : \mathbb{R} \to \mathbb{R}$ by $P(0) = 0$ and $P(x) = x^3 \sin(1/x)$ for $x \neq 0$. Prove that P is continuously differentiable on $\mathbb{R}$. Explain which bound differs from the discontinuous-derivative example in Lecture 8. Use the derivative definition at zero and the rule table elsewhere.